1. Clinical Question & Core Hypothesis

Clinical Question

PSA density (PSAD) reliably predicts disease reclassification, yet clinical discordance occurs — why does PSAD sometimes misclassify risk? The Ontologist suggests inflammation and EMT-associated loss of E-cadherin could decouple PSA production and gland architecture from true tumor invasive potential, producing PSAD–reclassification mismatches. 2) Tier 1 paths used: (a) PSAD → disease reclassification (direct clinical predictor), (b) inflammation → prostate cancer (modulates PSA production and gland remodeling), and (c) EMT → E-cadherin ↓ → worse prognosis (mechanism linking cell plasticity to aggressiveness). These provide a mechanistic chain linking stromal/inflammatory context and EMT to PSAD performance. 3) Tier 2 support: PSAD as a benign-versus-malignant discriminator and centrality of PSA levels underpin the biomarker framework; cross-cancer EMT links (e.g., lung) generalize the EMT→invasion concept; TP53 as a broad modifier suggests mutational context may further modulate outcomes. 4) Endpoint choice (PFI): disease reclassification is intended to predict progression risk; progression-free interval maps directly to that clinical consequence and is more sensitive than overall survival for earlier shifts in management. 5) Deliberate exclusions: I excluded Tier 3 exploratory paths (Barrett's esophagus, polyps, breast-tissue transitional-stage notes) because they are tangential to prostate PSAD biology. I also excluded overloading the core hypothesis with multiple genes or speculative mutational drivers (e.g., TP53) to preserve mechanistic focus; such factors are listed as secondary_modifiers for downstream testing?

Working Hypothesis

In localized prostate cancer, the predictive accuracy of PSA density (PSAD) for progression-free interval (PFI) is conditional on the tumor microenvironment and EMT status. Specifically, high stromal-to-tumor ratio (TSR) and low E-cadherin (CDH1) expression—indicative of an inflammatory/desmoplastic milieu and EMT activation—will attenuate the ability of PSAD to reflect true invasive tumor burden, leading to PSAD-based misclassification of progression risk. Conversely, tumors with low TSR and preserved CDH1 expression will show PSAD as a more faithful surrogate for tumor burden and progression risk. These effects should manifest as significant PSAD × TSR and PSAD × CDH1 interactions in time-to-progression analyses, with inferior discrimination (lower AUC) and higher misclassification rates for PSAD-based decisions in the high-TSR/low-CDH1 subgroup. If we incorporate TSR and CDH1 into a combined PSAD-TSR-CDH1 metric, predictive performance for PFI should improve over PSAD alone, reflecting a more accurate accounting of non-tumor PSA sources and EMT-driven tumor aggressiveness.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	PSA density, E-cadherin (CDH1)
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PSA density: PSA density (PSAD) reliably predicts disease reclassification, yet clinical discordance occurs — why does PSAD sometimes misclassify risk? The Ontologist suggests inflammation and EMT-associated loss of E-cadherin could decouple PSA production and gland architecture from true tumor invasive potential, producing PSAD–reclassification mismatches.

E-cadherin (CDH1): Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	inflammation	Supporting modifier from knowledge graph
Secondary	PSA levels	Supporting modifier from knowledge graph
Secondary	EMT program	Supporting modifier from knowledge graph
Secondary	TP53 mutation	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- inflammation: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- PSA levels: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.